REALTIME MONITORING FRAMEWORK AND ALERTING SYSTEM FOR UNDERGROUND COAL MINES USING ARDUINO UNO

PROJECT OVERVIEW

Designed and developed an IoT-enabled mine safety monitoring system that continuously monitors hazardous underground environmental conditions such as toxic gas leakage, high temperature, and fire. The system utilizes an **Arduino Uno**, multiple environmental sensors, Bluetooth communication, LCD display, and an audible alarm to provide real-time monitoring and immediate alerts, improving worker safety in underground coal mines.

PROBLEM STATEMENT

Underground coal mining environments expose workers to hazardous conditions including toxic gas leaks, excessive temperatures, and fire incidents. These dangers are difficult to detect manually and can lead to severe accidents. The project addresses this challenge by providing continuous environmental monitoring and instant alert generation to improve miner safety.

OBJECTIVES

- Continuously monitor underground environmental conditions.
- Detect hazardous gas leakage, abnormal temperature, and fire.
- Generate real-time audible and visual alerts when safety thresholds are exceeded.
- Enable wireless transmission of sensor data for remote monitoring.
- Improve the overall safety of underground mining operations.

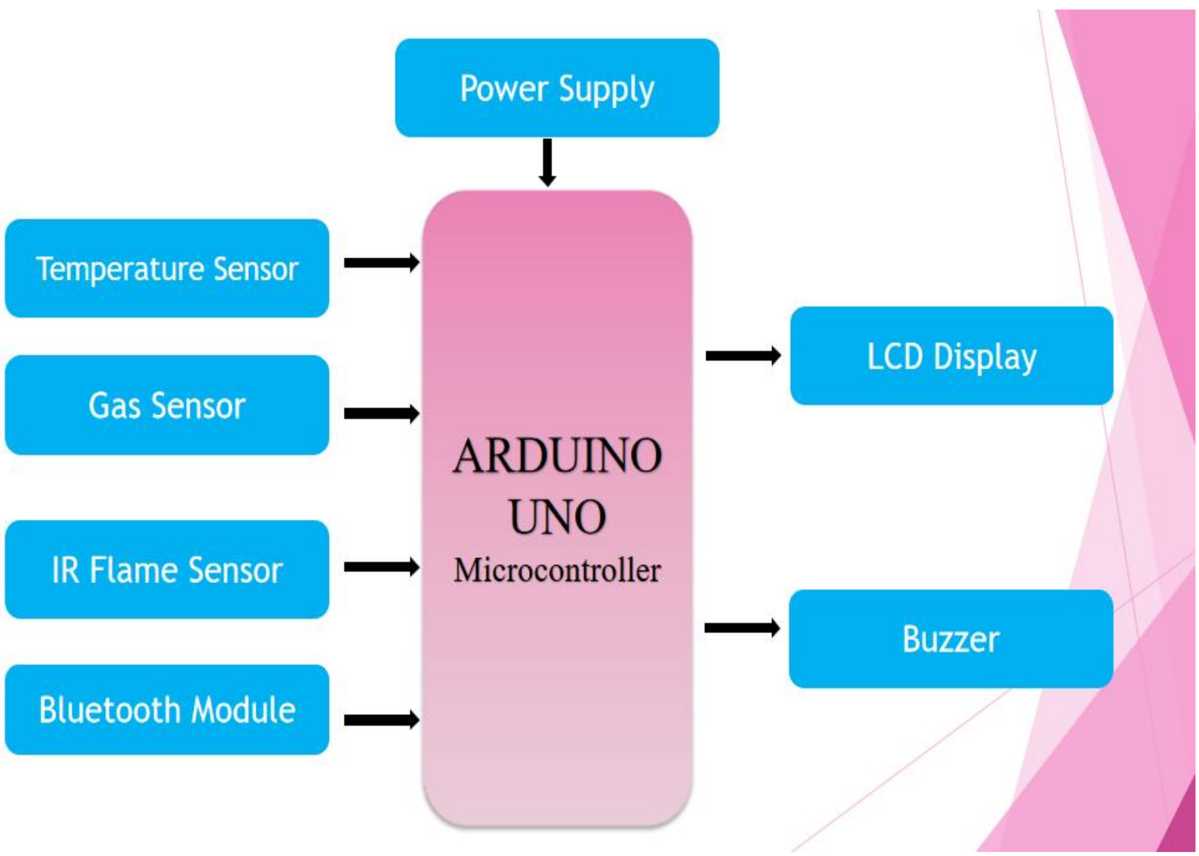
TECHNOLOGIES & COMPONENTS

- Arduino Uno
- LM35 Temperature Sensor
- MQ-2 Gas Sensor
- IR Flame Sensor
- HC-05 Bluetooth Module

- 16×2 LCD Display
- Piezo Buzzer
- Embedded C (Arduino IDE)
- IoT-Based Wireless Monitoring

SYSTEM WORKFLOW

1. The Arduino Uno continuously receives data from the temperature, gas, and flame sensors.
2. Sensor readings are processed and compared against predefined safety thresholds.
3. If hazardous conditions are detected, the buzzer sounds immediately and warning messages are displayed on the LCD.
4. Sensor data is transmitted wirelessly through the HC-05 Bluetooth module for remote monitoring.
5. Mine personnel can monitor environmental conditions in real time and respond quickly to emergency situations.



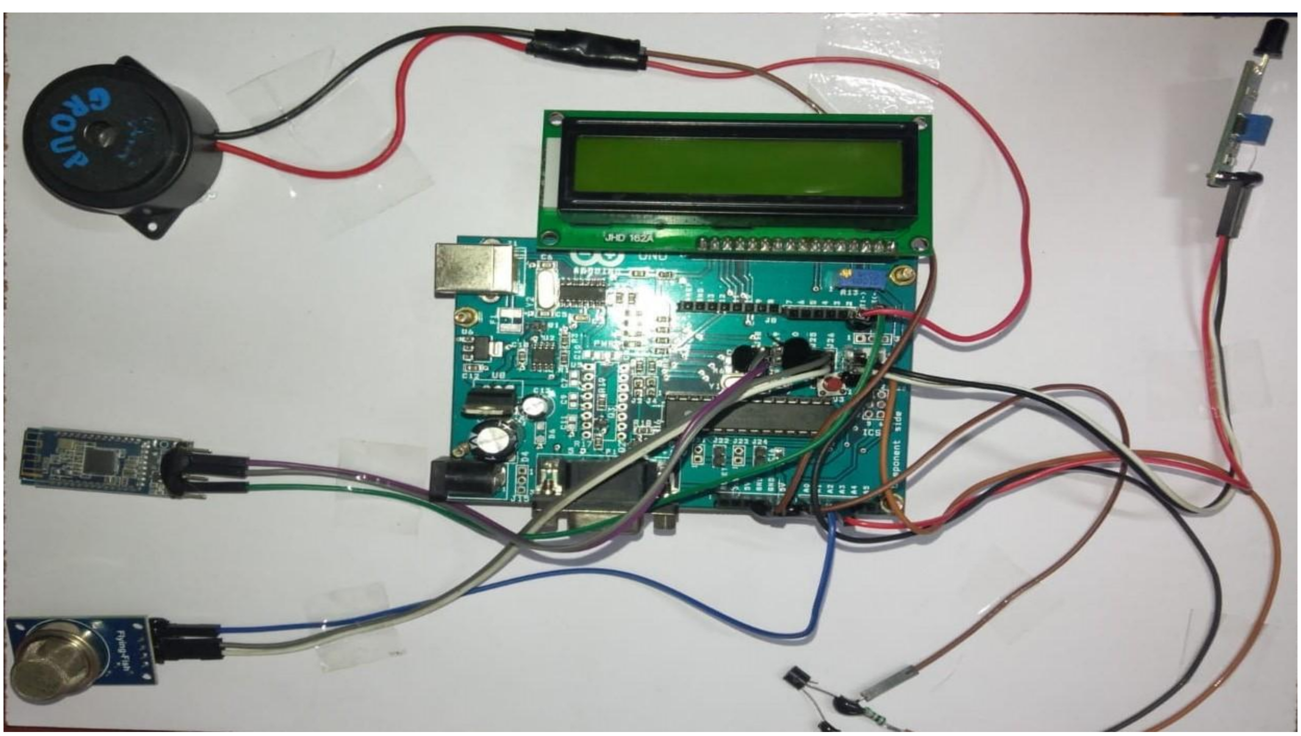
KEY FEATURES

- Real-time environmental monitoring
- Hazardous gas leak detection
- Temperature monitoring
- Fire detection using IR flame sensor
- Audible and visual emergency alerts
- Wireless Bluetooth communication
- Remote monitoring capability
- Low-cost embedded safety solution

RESULTS

The prototype successfully demonstrated:

- Accurate monitoring of underground temperature.
- Reliable hazardous gas detection using the MQ-2 sensor.
- Fast flame detection using the IR flame sensor.
- Successful wireless transmission of sensor readings through Bluetooth.
- Real-time monitoring with immediate alert generation for unsafe conditions.



APPLICATIONS

- Underground coal mine safety monitoring
- Industrial hazardous gas monitoring
- Fire detection and alert systems
- Worker safety and environmental monitoring
- Remote industrial monitoring solutions
- IoT-enabled safety systems for hazardous workplaces

FUTURE ENHANCEMENTS

Future improvements could include:

- Integration of additional sensors such as humidity, air quality, and seismic sensors.
- Autonomous emergency response mechanisms.
- Enhanced mobile application with improved visualization and analytics.
- Cloud-based IoT monitoring and data logging.
- Scalable deployment across different mining environments.

SKILLS DEMONSTRATED

- Embedded Systems Development
- Arduino Uno Programming
- IoT-Based Monitoring
- Sensor Interfacing
- Bluetooth Communication
- Embedded C Programming
- Real-Time Data Acquisition
- Safety Monitoring System Design
- Hardware Prototyping
- Alarm and Notification Systems